

A preliminary study of variations of the Changjiang Diluted Water between August of 1999 and 2006

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Abstract

A large area hypoxia has been already reported respectively by two interdisciplinary surveys off the Changjiang Estuary since summer of 1999 and 2006. The hypoxic zone shows distinct year-to-year variations. Observed oceanographic data are first analyzed and reveal a big difference for the Changjiang Diluted Water (CDW) between these two periods. These great changes are related to the tremendous reduction of the freshwater discharge and variations of wind fields between these two years. It is also found that the monthly mean intrusion of Kuroshio and its branches has increased in the northern East China Sea (ECS), but decreased in the southern ECS in August of 2006 as compared with 1999 on the base of general circulation models. Then, the Regional Ocean Modelling Systems is applied to the East China Sea to evaluate the contributions and relative importance of impacts from the river discharge, wind forcing and open boundary data. Our simulations reproduce the phenomena that more fresh water extends northeastward in 2006 and forms a negative SSS anomaly to the northeast of the river mouth as compared with 1999, which is consistent with observations. The five group numerical tests suggest that the wind forcing dominates the CDW variations followed by the Kuroshio and its branches. The study implies important roles played by hydrodynamic processes on the variability of hypoxic zone in the study areas.

Key words: Changjiang Diluted Water, variations, dynamics, hypoxia

1 Introduction

A large area of low dissolved oxygen (DO) bottom water has been a distinctive oceanic phenomenon in the Changjiang River Estuary and its adjacent areas in recent years (Li et al., 2002; Chen et al., 2007; Wei et al., 2007; Zhang et al., 2007; Zou et al., 2008). The hypoxic zone ($DO \leq 2.0$ mg/l) shows significant inter-annual variability. For example, according to two interdisciplinary cruises in summer of 1999 and 2006, the core region of low oxygen covers a large area outside of the Changjiang River Estuary during both years, however, the core region extended further northward in August, 2006 than the counterpart (Fig. 1). The formation and evolution of hypoxia are well documented and is believed to result from combination of physical, biological and chemical processes, among which vertical stratification and water movement are of the first importance (Officer et al., 1984; Kuo et al., 1987; Stanley et al., 1992; Park et al., 1996; Diaz, 2001; Ra-

balais et al., 2002; Grantham et al., 2004; Yin et al., 2004). In order to understand the roles and influences of hydrodynamic processes on the formation and evolution of the hypoxia in the study area, the first step is to explore the variability of the hydrodynamic processes themselves.

The extension of the Changjiang Diluted Water (CDW) is worthy of first consideration due to its distinctive contributions to nutrients and water transportation (Wang et al., 2002). In the East China Sea, riverine nutrients from the Changjiang River account for a large percent of total nutrients (Tian et al., 1993; Pu et al., 2001), but Kuroshio intermediate water is also an important external source, especially for phosphorus (Chen et al., 1999). In summer, the mixture of cross-shelf Kuroshio Water (KW) and Taiwan Strait Water (TSW), which is also famous as the Taiwan Warm Current (TWC), can reach the submerged river valley outside the Changjiang River mouth (Weng et al., 1984), whose destination finally connects tightly

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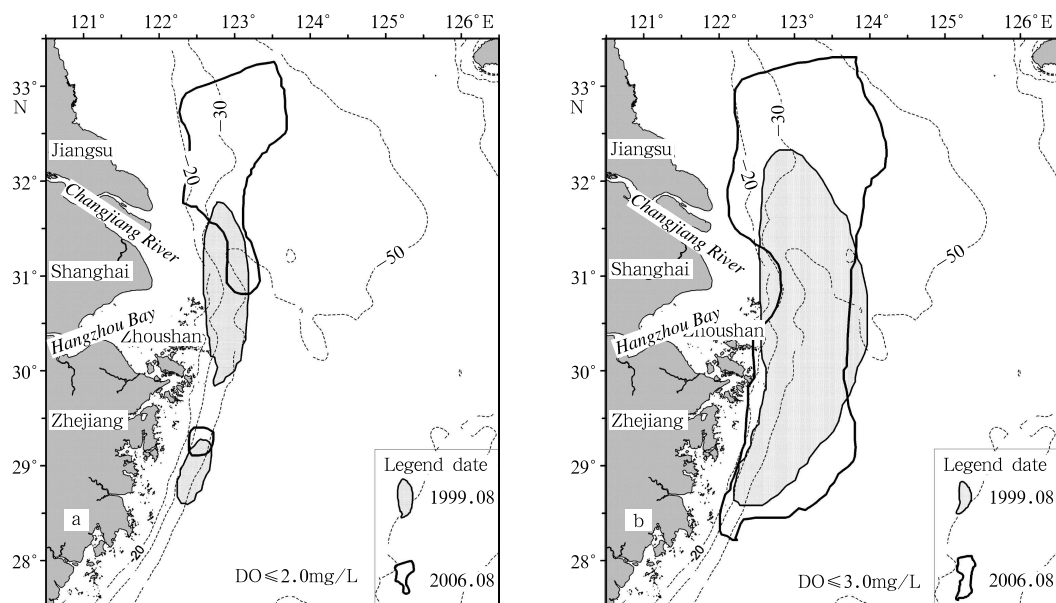


Fig.1. Comparisons of bottom low oxygen region in the studying areas in summer between 1999 and 2006. a. $DO \leq 2.0$ mg/l; b. $DO \leq 3.0$ mg/l. The shading area is digitized from Li et al.(2002, Fig 1); the thick contours are in situ observations in Aug. 2006.

with the spreading of the CDW. Due to the important impacts of the CDW on the transportation of nutrients and further on the distribution of eutrophication and phytoplankton (Ning et al., 2004), the link can be established between the CDW extension and hypoxia locations. This link is essentially true during periods when the CDW varies greatly.

In summer, the most pronounced phenomenon of the CDW is its northeastward or eastward spreading instead of southward spreading as is expected due to the coriolis effects (Su, 2001). There are many literatures working on this issue from the aspects of field data analysis (Mao et al., 1963; Beardsley et al., 1985; Gu, 1985; Wang et al., 1986; Le, 1989) and numerical modeling (Zhao, 1991; Zhu et al., 1997; Zhu et al., 1997; Zhu et al., 1998; Zhu et al., 1999; Liao et al., 2001; Bai et al., 2003; Chen et al., 2008). Zhao et al. (1991) gave a thorough review of the driving mechanism of the spreading of the CDW, suggesting that three major forcing are responsible for the phenomenon: (1) the Changjiang River run off; (2) Taiwan Warm Current; (3) wind force. However, it is still unknown how these factors affected the extension of the CDW in the specified month in 1999 and 2006. Which factor played a dominant role on the spreading of CDW.

This study first analyzed comparatively in situ measurements collected during two cruises in Aug 1999

and 2006, especially the temperature, salinity and dissolved oxygen distribution, and then adjusted a numerical model to investigate the CDW variability and impacts from the three driving force. This paper aims to make tentative efforts to understand the relationship between the hydrodynamics and hypoxia variability and try to establish a practical method to predict the hypoxic zone migration in the study area.

2 Data and methods

2.1 Data

The study area is shown in Fig. 1. The data were collected from two interdisciplinary cruises on the scientific R/V Beidou during 20–29 August 1999 and 15–31 August 2006. The most significant phenomenon is the hypoxic zone in the bottom dissolved oxygen during both of these two periods. The main hypoxic zone distributes outside the mouth of Changjiang River, while the other part is much smaller along the coast of the Zhejiang Province. The distribution of bottom dissolved oxygen shows big discrepancy between the two years. The core region of hypoxia in 2006 spreading northward far away than that of 1999. The migrations of the low oxygen region exceed one degree in the meridian direction and can further be demonstrated by the DO isolines of 3 mg/l. The comparison also indicates that the hypoxic zone increases by a

large percent between the two years.

2.2 Model configuration

The three-dimension primitive equation model Regional Ocean Modeling Systems (ROMS 3.1) (Haidvogel et al., 2000; Ezer et al., 2002; Shchepetkin et al., 2003; Shchepetkin et al., 2005) is used in this paper. The model includes the Bohai Sea, the Yellow Sea and the East China Sea covering 24° – 41° N, $117^{\circ}30'$ – $131^{\circ}30'$ E, in which the north, east and south edge are open boundaries according to the realistic bathymetry, while the west boundary is closed for the same reason. The horizontal resolution is $5' \times 5'$ in each direction. There are 30 terrains following (S-coordinates) (Song et al., 1994) layers in the vertical direction; the parameter θ_s is 5.0 and θ_b is 0.4 so that the model can resolve finely both the surface and bottom boundary layers. Those areas deeper than 2 500 m are cut to 2 500 m to concentrate on the shelf regions but without affecting the Kuroshio effects. The sub-grid vertical mixing uses KPP scheme developed by Large et al (1994) and mixing at the bottom boundary is improved by Durski et al.(2004). The horizontal mixing and diffusion are calculated along S-coordinates layers, where the viscosity coefficient is $5 \text{ m}^2/\text{s}$ for the velocity and the diffusion coefficient is $100 \text{ m}^2/\text{s}$ for temperature and salinity respectively.

To simplify the problem, the model takes only the M_2 tides into consideration, but is sufficient for the model to enhance the mixing processes reasonably because of dominant role of M_2 component in the study area. The time step is 207 s, which results in 72 times step over an M_2 cycle and makes it easy to remove tidal signals. In order to supply the M_2 tidal elevation and phase, a large domain Hamburg Shelf Ocean Model (HAMSOM) (Backhaus, 1985; Huang et al., 1999) is applied at first and the tidal boundary data are prepared for ROMS with a similar method mentioned by Zhou et al. (2009). The open boundary uses Flather conditions (1976) for barotropic velocity and Chapman condition (Chapman, 1985) for sea surface elevation to include external M_2 forcing, while other parameters use radiation conditions. The technique of nudging is used at all open boundaries for temperature, salinity and baroclinic currents: the nudging time is 30 d, 30 d and 180 d for the outflow while 1/3 d, 1/3 d and 2 d for inflow. The model also includes the Changjiang River as a point source in the lateral boundary. The annual mean run-off of the Changjiang River is $2.8 \times 10^3 \text{ m}^3/\text{s}$, but has strong seasonality and

interannual variations (Chen et al., 2001), especially for the year of 1999 and 2006. The detail will be presented in the following section 2.3.

The heat flux and fresh water flux are prescribed at the surface boundary extracted from the Comprehensive Ocean-Atmosphere Data Set (COADS) (Da Silva et al., 1994). The model was initialized from the temperature, salinity field from the multi-year mean dataset called Atlas of the Bohai Sea, the Yellow Sea and the East China Sea (Chen, 1992); The boundary data like temperature, salinity, velocity and sea surface elevation are multi-year mean (1979-2003) from the Hybrid Coordinate Ocean Model (HYCOM, <http://hycom.rsmas.miami.edu/dataserver/>) (Bleck, 2002). The model ran continuously for 4 years and achieved fundamentally dynamic equilibrium (hereafter ref to as control run), and the results from the last year are used in the following context as climatological fields and results from the third year are as initializations for sensitivity cases.

2.3 Sensitivity tests

After the climatological simulation, five tests are conducted respectively to investigate the individual and combined effects on the Changjiang Diluted Water extension from the river run-off, wind and open boundary volumes (OBS). Each test contains two comparative runs driven by monthly forcing of 1999 and 2006, which is listed in detail in Table 1. Sensitivity tests warm start from the climatological run of the 3rd model year. In each case, the specified year forcing replaces the corresponding forcing of the control run. For example, the runoff of 1999 and 2006 will take place of the multi-year mean runoff in the first case.

Table 1. Design for sensitivity tests

Test	Run off	Wind	OBS
Case 1	2006 vs 1999	–	–
Case 2	–	2006 vs 1999	–
Case 3	–	–	2006 vs 1999
Case 4	2006 vs 1999	2006 vs 1999	–
Case 5	2006 vs 1999	2006 vs 1999	2006 vs 1999

The run off of the Changjiang River during the summer of 1999 and 2006 has remarkable contrast (Fig. 2). It is very drought in summertime of 2006 while the counterpart is in overwhelming flood. Wind data come from Blend Sea Winds (Zhang et al., 2006) and then stress is calculated using Large and Pond method (Large et al., 1981). Figure 3 presents the

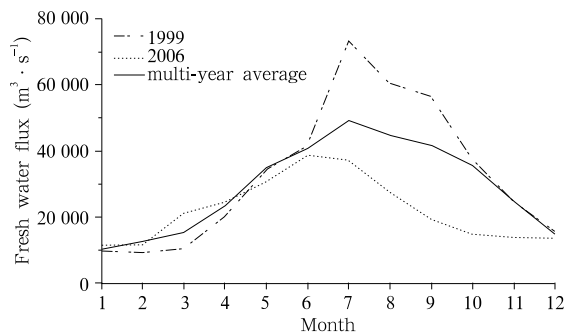


Fig.2. Comparisons of monthly mean run off at Datong hydrological station of the Changjiang River among multi-year average, the year of 1999 and 2006.

wind vectors and shows big changes between the year 1999 and 2006. The boundary velocity and sea surface elevation are provided by HYCOM model results of corresponding years, from which the volume transport in the southern edge is examined and listed in Table 2. The volume comparison at the southern boundary shows that the Kuroshio in the east of Taiwan increased by $2 S_v$ ($1 S_v=10^6 \text{ m}^3/\text{s}$) in summer 2006 comparing with 1999. Before input into the model, the volume transports at all three boundaries are adjusted to keep volume conservative. The changes in the southern boundary represent the impacts resulting from the variability of Kuroshio and TSW.

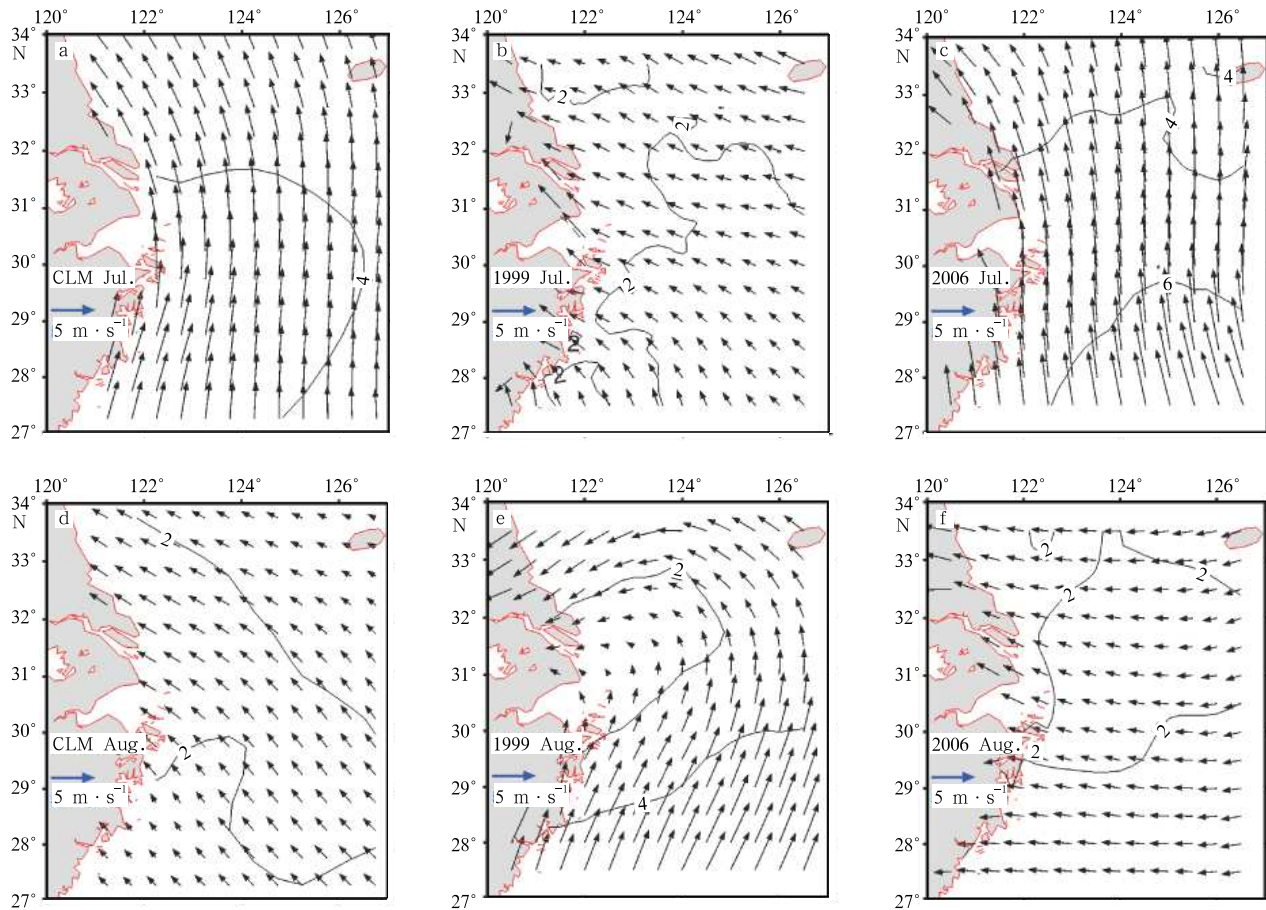


Fig.3. Comparisons of wind fields in July and August among multi-year average (CLM), the year of 1999 and 2006.

Table 2. Volume transportation at the southern boundary (unit: S_v)

Location	Climatology 1999 2006		
Taiwan Strait in July	3.0	2.6	1.2
Taiwan to 124°E (Kuroshio) in July	30.0	24.2	24.1
Taiwan Strait in August	2.1	1.6	1.5
Taiwan to 124°E (Kuroshio) in August	23.8	24.7	27.1

2.4 Model validation

The simulated results of the last year of the control run are verified to make sure that the model can reproduce the spreading of the Changjiang Diluted Water and is in consistent with the observations. The multi-year averaged sea surface salinity is digitized from the Atlas of the Bohai Sea, and the Huanghai

Sea and the East China Sea (Chen, 1992) is used as observations (Fig. 4). The river plume pattern has been identified by contours of 26, 30 and 31 as Mao et al. (1963) and Beardsley et al. (1985) did. The

tongue-like pattern is very significant in July and August when the isolines 26 show a stronger northeastward branch and a weaker southeastward branch. The extending distance can be further represented by

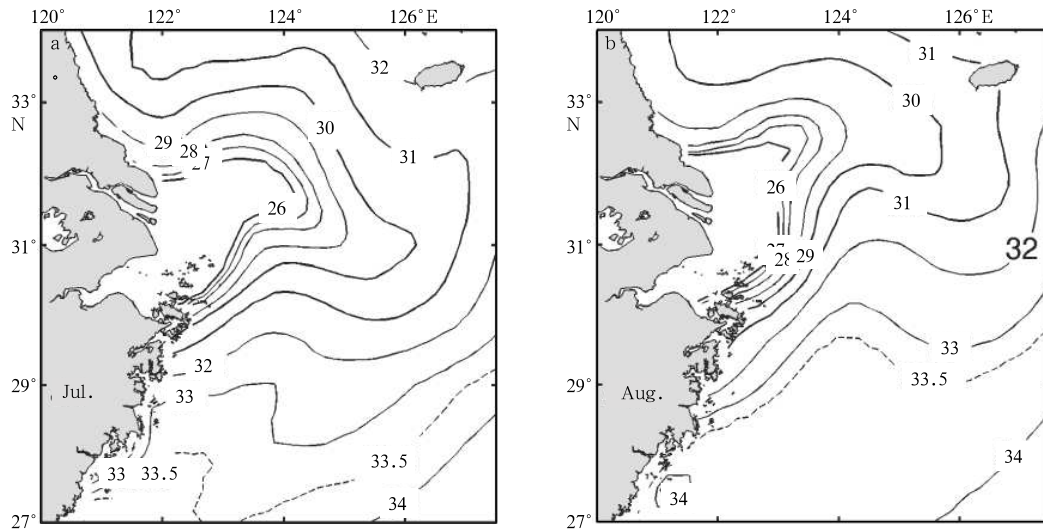


Fig.4. Multi-year averaged sea surface salinity in July and August in the studying areas according to Chen et al. (1992).

isolines 30, which distinguish the northeast branch farther with sparse gradient, but the southeast branch closer with high gradient. It can be also found that the patterns in these two months are different from each other as follows: (1) The northward extension of the CDW goes farther in August than in July according to isolines 26, but eastward extension is stronger in July than in August; (2) The CDW shows a notable trend of extending to the Cheju Island in August according to the end tip of the isolines 31, but it

turns to the southeast in the west of the Cheju Island in July; (3) The northward incursion of saline water is much stronger in the southern region of the study areas in August than in July, which implies the intensity difference of Kuroshio and TSW during these periods.

The model can catch characteristics mentioned above in a great extent as shown in Fig. 5 and so the model performance was convincing.

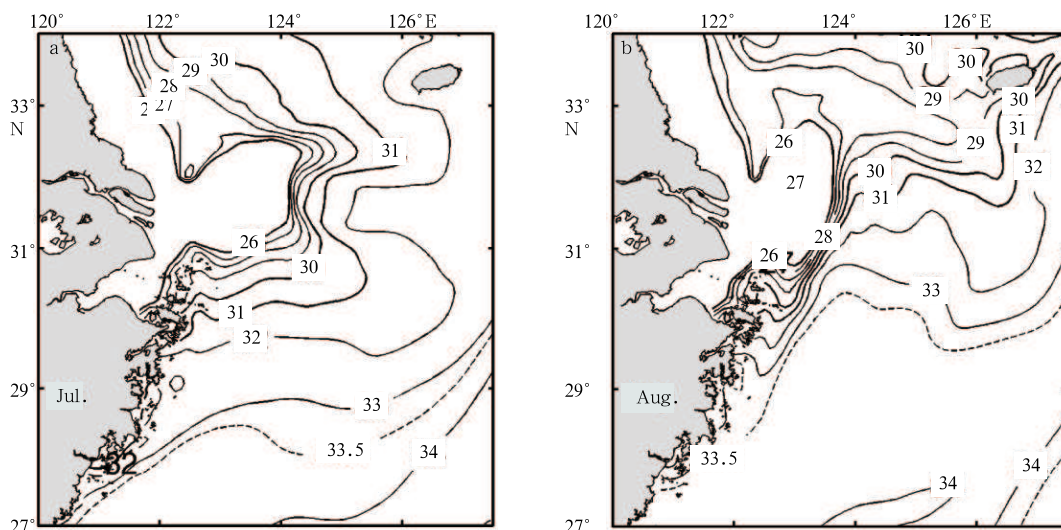


Fig.5. Simulated sea surface salinity in July and August from the last year of the control run.

3 Results

3.1 CDW observations and variations

The measured sea surface salinity (SSS) in August of 1999 and 2006 are given in Fig. 6 and their difference is also calculated by subtracting the values in 1999 from that in 2006. The SSS is significantly lower in 1999 than the multi-year average (Fig. 4) which is a correspondence induced by the flood situation in the drainage basin of the Changjiang River in that season, while it is more saline in 2006 due to the tremendous reduction of the river discharge. Their patterns show great discrepancy compared with those

shown in Fig. 4. In 1999, the CDW is very intensive; the southward branch appears to be particularly stronger than the northeastward part and can arrive at the 100 m isobaths. On the contrary, the CDW is pretty weak and confined to the near-shore region in August 2006. One striking feature in this year is that a block of tongue-like saline water separates the CDW into two independent regions with fresh water in the north and south direction respectively. However, if compared with the flood year 1999, the SSS in the northern region of river mouth in drought year 2006 is lower. Therefore an unexpected negative SSS difference exists in the above areas (Fig. 6 right).

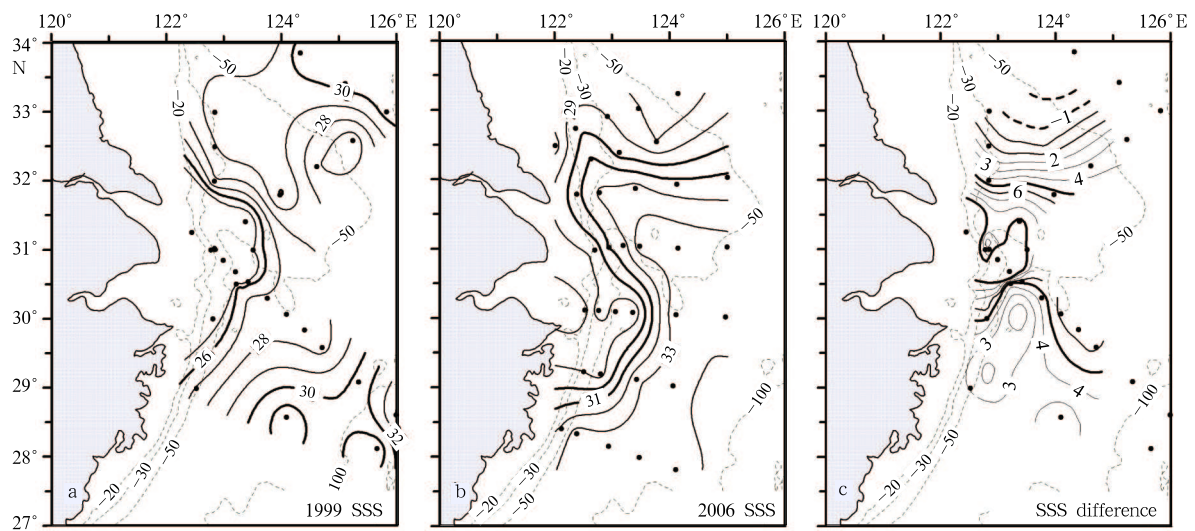


Fig.6. Comparisons of observed sea surface salinity in August of 1999 and 2006. The difference is calculated after they are interpolated into the same grids. The dots indicate the measuring stations.

The above observed inter-annual spatial patterns raised following questions: how did the negative SSS difference form? Whether there is connection between the negative SSS difference and the northward migration of the hypoxic zone in 2006. If so, how do they connect? The numerical model is a necessity to answer these questions.

3.2 Modeled CDW variations

The model ROMS is adapted to conducting five sensitivity tests. The first three tests are designed to evaluate impacts from three driving forcing: river run off, wind and open boundary transport respectively. The simulated SSS in August of 1999 and 2006 are compared to identify the role of the individual forcing contribution to the extension of the CDW.

In the first case the difference of river run off doesn't make a big difference to the CDW pattern be-

tween the two years (Fig. 7 left). The main changes are limited to the west of 124°E. Both years show their fresh water spreading mainly to the northeast region. There is only slightly difference for the southeastward branch of the CDW which goes far distance in 1999. The first case can't explain the different spatial distribution of the CDW observed in August 1999 and August 2006.

Wind variations resulted in CDW extension very different in the second case (Fig. 7 middle). The southeastward branch is dominant instead of the northeastward branch in 1999. By contrast, the northeastward branch is still the most typical feature in 2006. So the wind effects need to be further investigated and it will be further presented in the fourth and fifth cases.

The third case is designed to understand contributions from open boundary data which stand for the

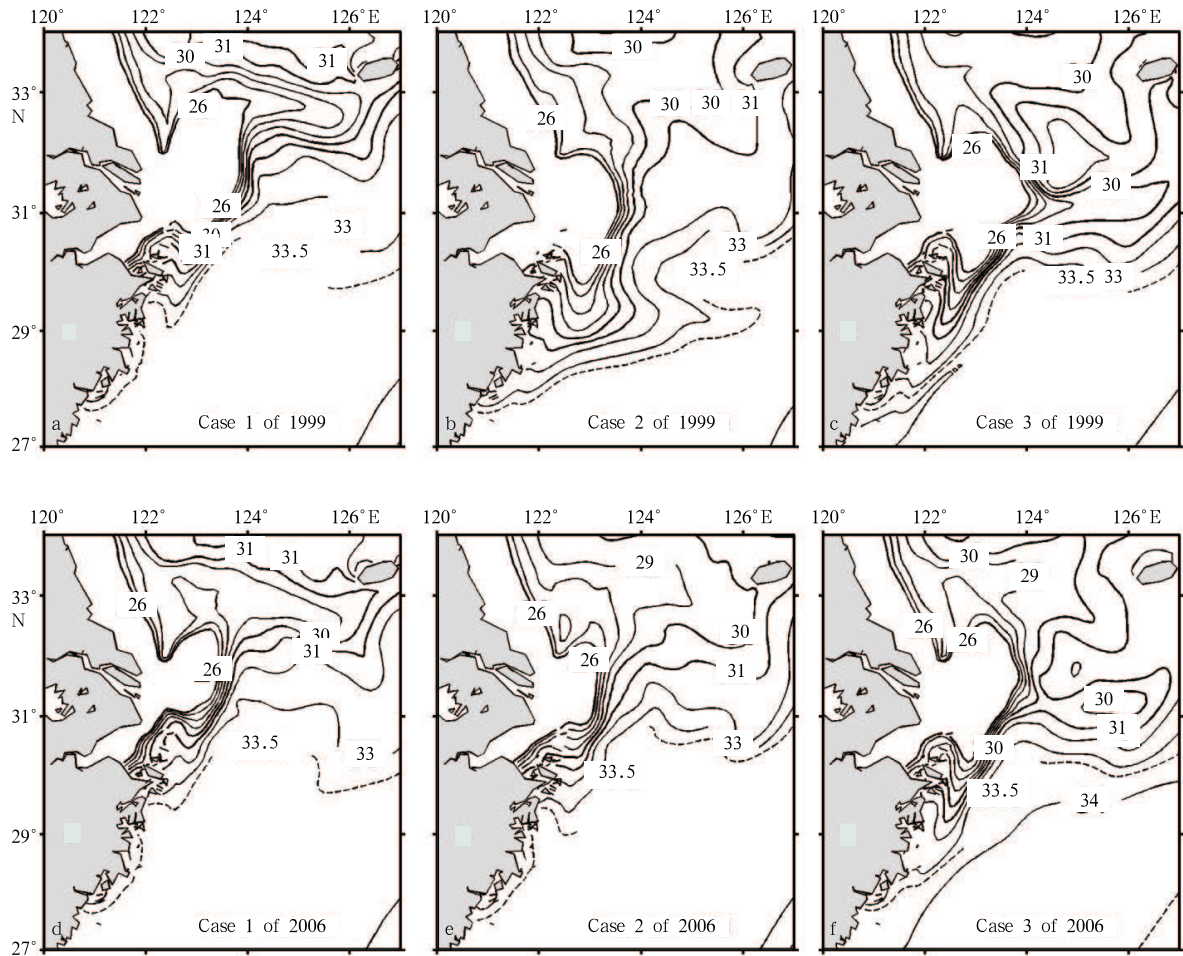


Fig.7. Distributions of simulated sea surface salinity in August of 1999 and 2006 from case 1-3. Contour interval is homogeneous at 1 PSU indicated by solid line, while the salinity larger than 33.5 are plotted additionally as dashed lines at 1 PSU interval, too.

impacts from the Kuroshio and TSW (Fig. 7 right). As compared with the first two cases, the eastward extension is enhanced in the third case for both years, but there are some differences between these two years. The eastward and southward extension is relatively strong in 1999 while the eastward and northward extension is more distinctive in 2006. In addition, the water in the southern region of the study areas is much saline in 2006 than in 1999, which looks similar to the observations in Fig. 6 (left two) and are not reproduced by the first two cases.

The fourth and fifth sensitivity tests are deployed to understand the combined effects of river discharge and wind, and all of the three forcing on the spreading of the CDW. A new variable ΔSSS is defined as $\Delta SSS = SSS_{2006} - SSS_{1999}$ to compare the difference SSS between 1999 and 2006 caused by different factors, among which SSS_{2006} delegates SSS in August 2006 and SSS_{1999} delegates SSS in August 1999. The re-

sults of July and August from these two cases are listed along with the first case results in Fig. 8. The runoff variations produce a block of positive ΔSSS in July as well as in August. But both of them mismatch completely with observations in Fig. 6 (right). If wind forcing is further taken into consideration like in the fourth case, the results appear better. It can be clearly seen that besides a positive ΔSSS , there is a negative ΔSSS in the east of the river mouth in July and in the northeastern region in August. The ΔSSS pattern in the latter case looks consistent with the measurements which suggests the important roles played by wind forcing variations. The final case gives us more information about the contributions from the Kuroshio and TSW, which on one hand push less saline water from the river estuary far more northward in 2006 and on the other hand let more fresh water move eastward far away in 1999. So the negative ΔSSS is enhanced and enlarged. Moreover, the positive ΔSSS was en-

larged in the final case than in the fourth case, too.

3.3 Modeled currents fields variations

The flow pattern is also presented (Fig. 9) and

gives more details about how the water properties were changed by advection between the two periods. The flow variation is defined using a similar method as before by simply replacing SSS with velocity

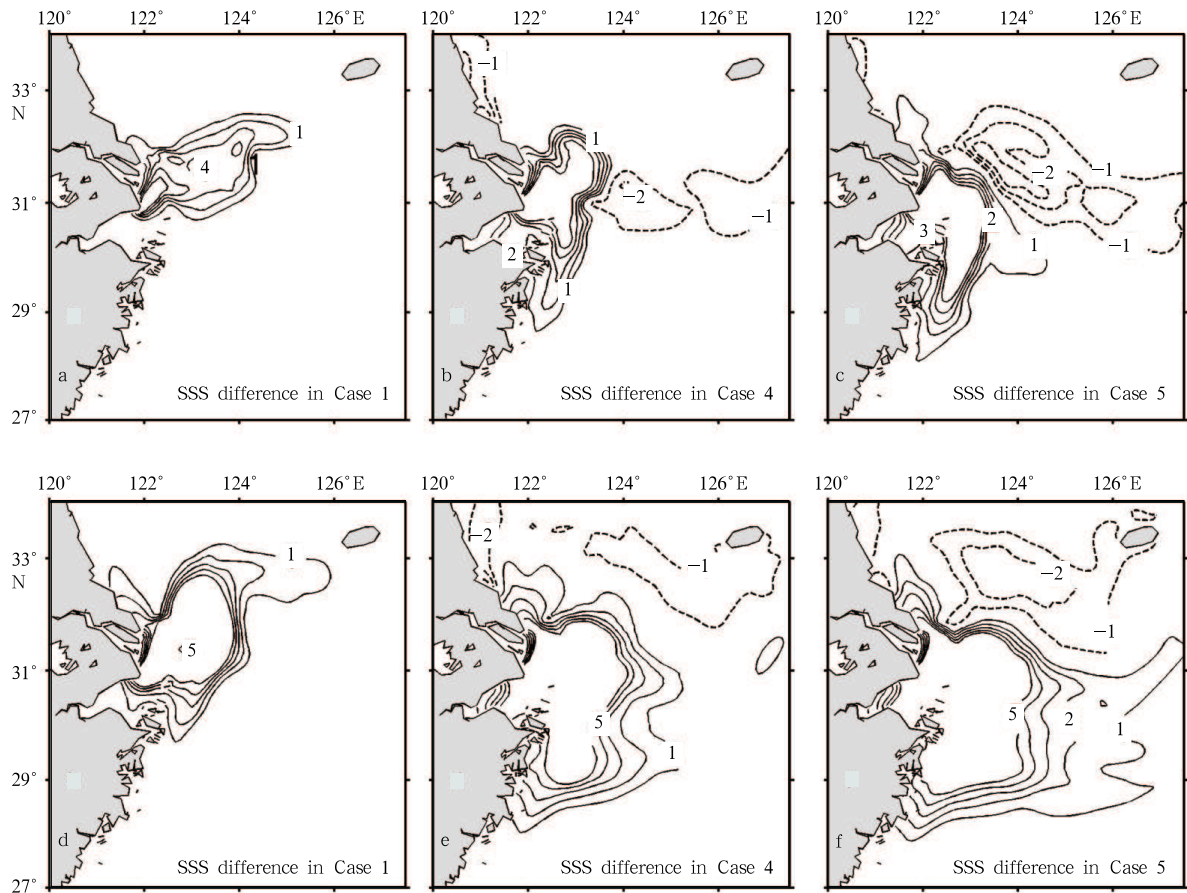


Fig.8. Distributions of sea surface salinity difference in July (top) and August (bottom) from case 1, 4 and 5. Positive differences stand for more saline situation in 2006 as shown by solid contours, while negative differences are denoted by dashed lines.

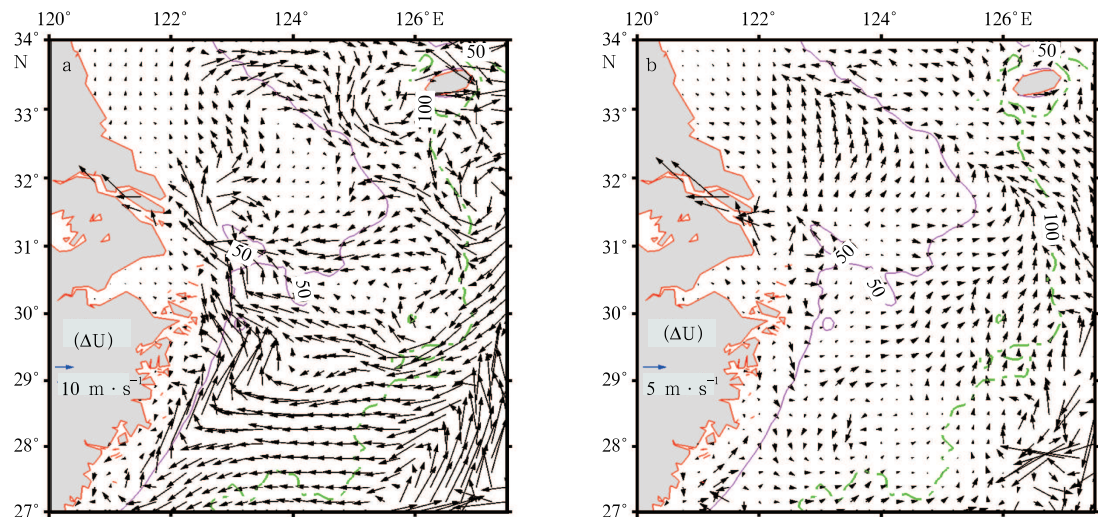


Fig.9. Distributions of simulated velocity difference at the surface (a) and bottom layers (b) in August from case 5. Solid lines indicate isobaths of 50 m and 100 m.

U : $\Delta U = U_{2006} - U_{1999}$, the U_{1999} , U_{2006} are current vectors in the two years and are calculated in the north and east direction respectively. Only the results in August are shown.

The surface ΔU suggests that the relative weakness of the southeast branch of the CDW in summer of 2006 is induced by the expansion of the offshore saline water. On the other hand, northeastward transportation is enhanced from the river mouth to northern region of the study areas in 2006 comparing with 1999. So in the eastern and southeastern region of the river mouth, the westward and northwestward flow anomaly is positive, while from the river mouth to the northern region, the northward and northwestward flow anomaly is positive in both of the surface layer and bottom layer. Further, in the southern region of 28°N, it has a different situation compared with other regions, the Kuroshio branches are weak in summer of 2006 than in 1999 which indicates that this region is dominated by the TSW instead of the Kuroshio as presented in Table 2.

4 Discussion and conclusions

Through ROMS model, five cases of comparative studies are conducted on the variability of the Changjiang Diluted Water caused by the Changjiang River runoff, wind field and the mixture water of the Kuroshio and TSW. Their dynamic effects are discussed on the migration of the hypoxic zone between 1999 and 2006. The studies suggest that physical factors are not only an important factor in the formation of low oxygen phenomenon (Stanley and Nixon, 1992), but also potential causes of their year to year variability. For the specified year of 1999 and 2006, the variation is mainly related to the wind forcing differences and is secondly due to is the difference of the intensity of the Kuroshio and TSW. The interaction between wind fields and the Kuroshio and TSW contribute to the northward moving of the CDW in 2006, or in other word, farther southward moving of the CDW in 1999. But the influence of river discharge is relatively small and confined to the near-shore region of the river mouth.

This paper is based on certain assumptions that physical processes link with biological and chemical processes by water and nutrients transportation which is essential for the formation of the hypoxia. As mentioned in many literatures, the extension of the Changjiang Diluted Water plays an important role in

riverine nutrients distribution. Moreover, since the saline Taiwan Warm Current as a result of mixture between the Kuroshio and TSW can reach the submerged river valley outside the Changjiang River mouth, it can be deduced that the CDW will have effects on the transportation of the external nutrients from the Kuroshio and its branches, too. These nutrients are fundamental for the fast and abundant production of phytoplankton, which in turn will consume dissolved oxygen during their decomposition. The hypoxia will occur if the oxygen consumption exceeding the replenishment. The simulated CDW variations agree well with the hypoxic zone migrations between 1999 and 2006 and prove the rationality of our assumptions. This study demonstrates the importance of physical processes besides biological, chemical process for the formation and evolution of hypoxia.

In the context, this paper concentrates only on the hydro-dynamic effects on the hypoxia, but does not suggest that dynamics factor is more important at any time than other factor accounting for the hypoxia. Several other dynamics factors need to pay attention to, such as the Yellow Sea Coast Current. They are not investigated in this paper due to the lack of measurements. On the other hand, the resuspended organic matter may also be important for the sake of decay and oxygen consumption, which require a combination of sediment dynamics but beyond the scope of this study. So, the relative importance of these two processes still need pay more attention, which helps us to understand in a comprehensive way on the primary mechanism of formation and evolution of hypoxia events.

There are several other literatures presenting hypoxia phenomenon outside the Changjiang River Estuary recently (Chen et al., 2007; Wei et al., 2007; Zou et al., 2008), but no systematic plan has been deployed to monitor the region yet. Due to the lack of measurements, the data sampled in the typical year of 1999 and 2006 are analyzed as the first approach to investigate the relationship between the water transportation and the hypoxic zone migration. According to this study, it can be deduced that the diagnostic of the CDW spreading is a short-cut to detect the hypoxic zone variations, since forcing data like winds, Kuroshio transportation and river discharge as well as other forcing like heat and water flux are on the lists of regular observations and can be easily obtained.

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